MDM2 SCREENING PIPELINE · ROOT-CAUSE REVIEW

Why the model looks confident — but picks compounds outside its range

A visual explanation for a non-specialist, built from the code, the saved metrics, and the cached 154,587-row screening table.

645 training compounds → 154,587 library compounds
0 of the top 100 are inside the applicability domain.

Prepared 2026-10-26 · local review · cached run: 154,587 rows (canonical 154,647; 60-row difference)

THE SINGLE POINT

The model is solving a different problem than the one you need

It is not mainly one wrong line of code. It is a mismatch between the chemistry it learned and the chemistry it is asked to rank — made worse by selecting the most extreme predictions.

It learned

645 MDM2 compounds from a limited chemical neighborhood. 204 Murcko scaffolds; top 10 cover 39%.

It is asked

Rank 154,587 diverse library compounds. 99.4% are below the 0.40 similarity cut; 99.94% are below the strict 0.52 cut.

It does

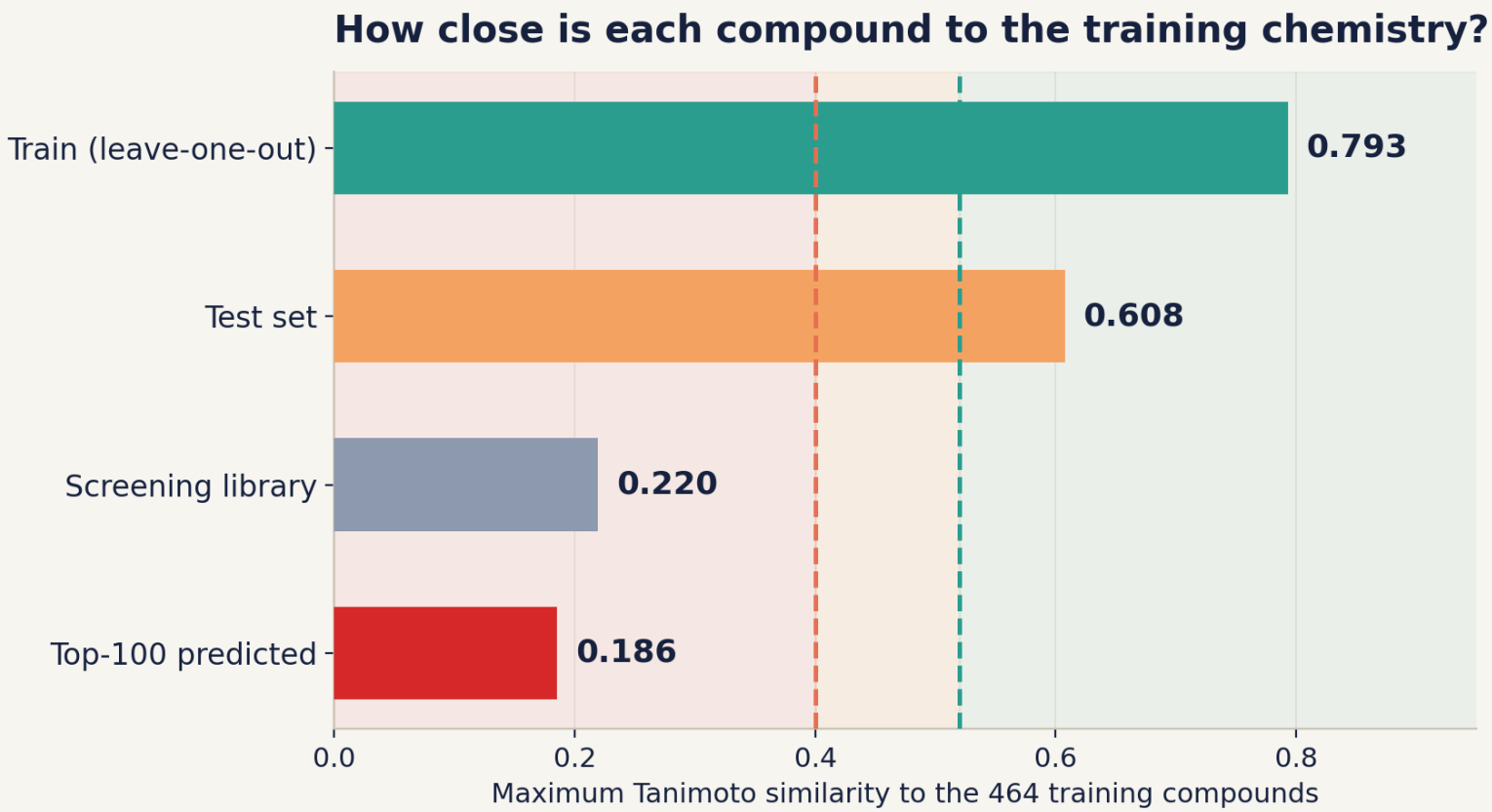
Takes the top predicted pIC50 values. Those values are highest where the model has the least evidence.

A model can output a number everywhere. That does not mean it has evidence everywhere.

The algorithm is secondary. The data coverage and the selection rule are primary.

Similarity collapses outside the training set

Max Tanimoto similarity to the 464 training compounds. Higher = more evidence; lower = more extrapolation.



- Train LOO mean: 0.793
- Test mean: 0.608
- Library mean: 0.220
- Top-100 mean: 0.186
- Library median: 0.217
- Top-100 max: 0.239
- Dashed lines: 0.40 AD / 0.52 strict

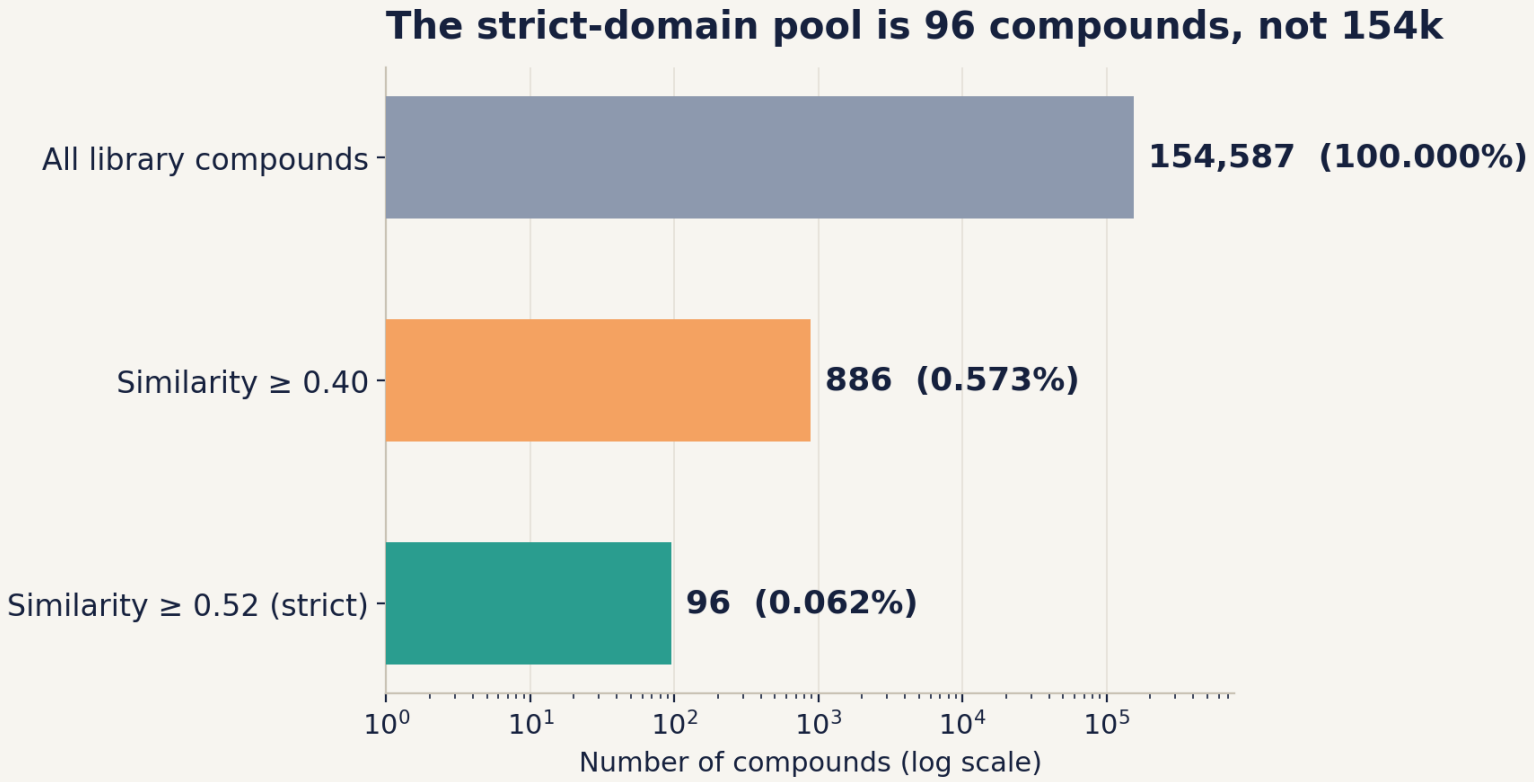
Only 0.57% of the library is inside the 0.40 cut.
Only 0.062% is inside the strict 0.52 cut.

Red zone = extrapolation (<0.40) · amber = caution (0.40–0.52) · green = in domain (≥0.52).

THE REAL SEARCH SPACE

The strict-domain pool is only 96 compounds

The library is not a 154k-choice buffet. Strict applicability-domain filtering leaves almost nothing to rank.



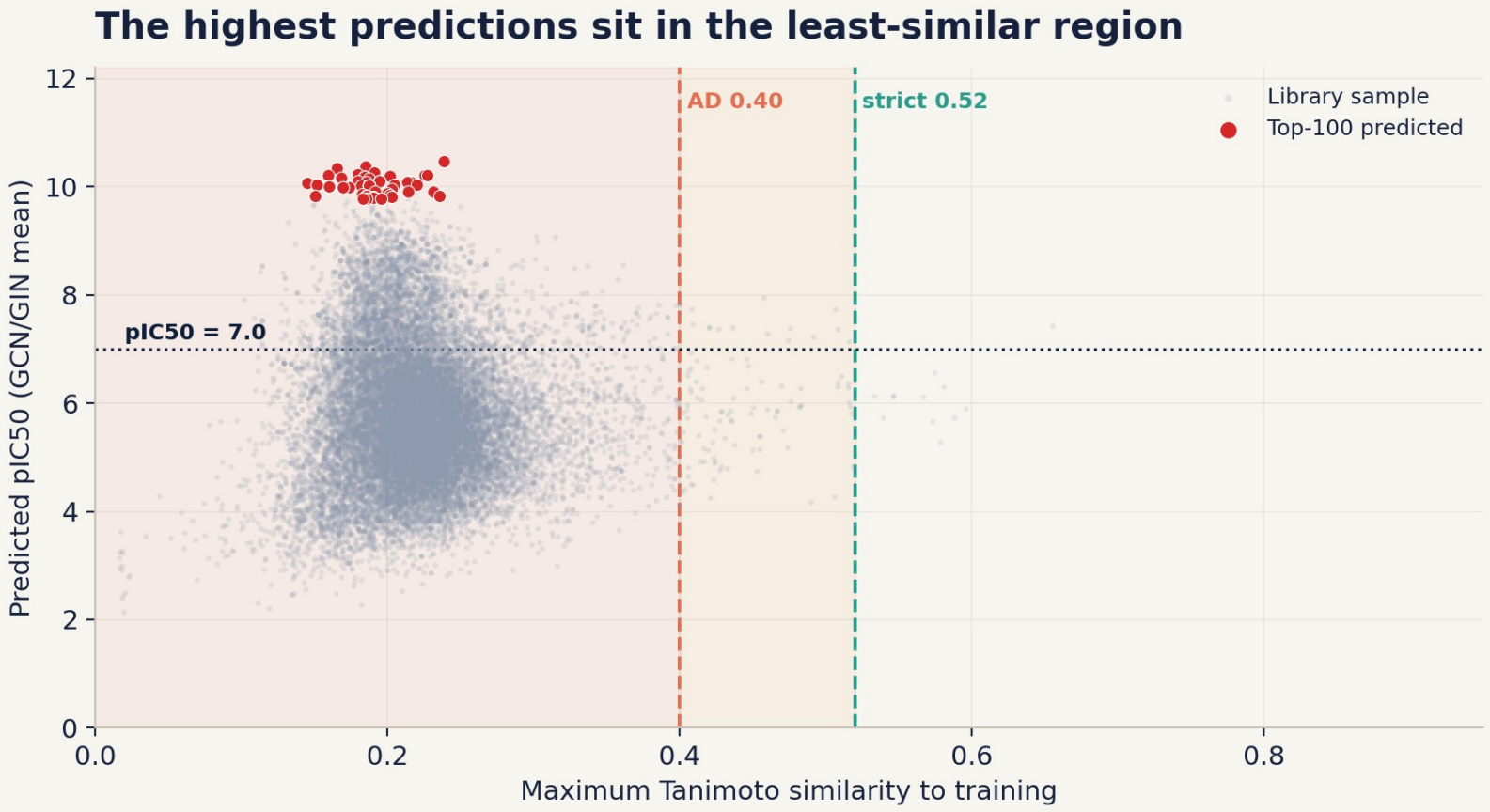
- 154,587 total library compounds
- 886 pass similarity 0.40 (0.57%)
- 96 pass strict 0.52 (0.062%)
- 0 of the top 100 pass 0.40
- 0 of the top 100 pass 0.52

If active support and model agreement are also required, the pool may be empty. An empty shortlist is a valid scientific result.

THE SMOKING GUN

Top predictions are the least similar compounds

Predicted pIC50 versus maximum similarity to training. The red points are the top 100 by prediction.



Top-100 signal

Mean similarity 0.186
Library median 0.217
All 100 outside AD 0.40

What this means

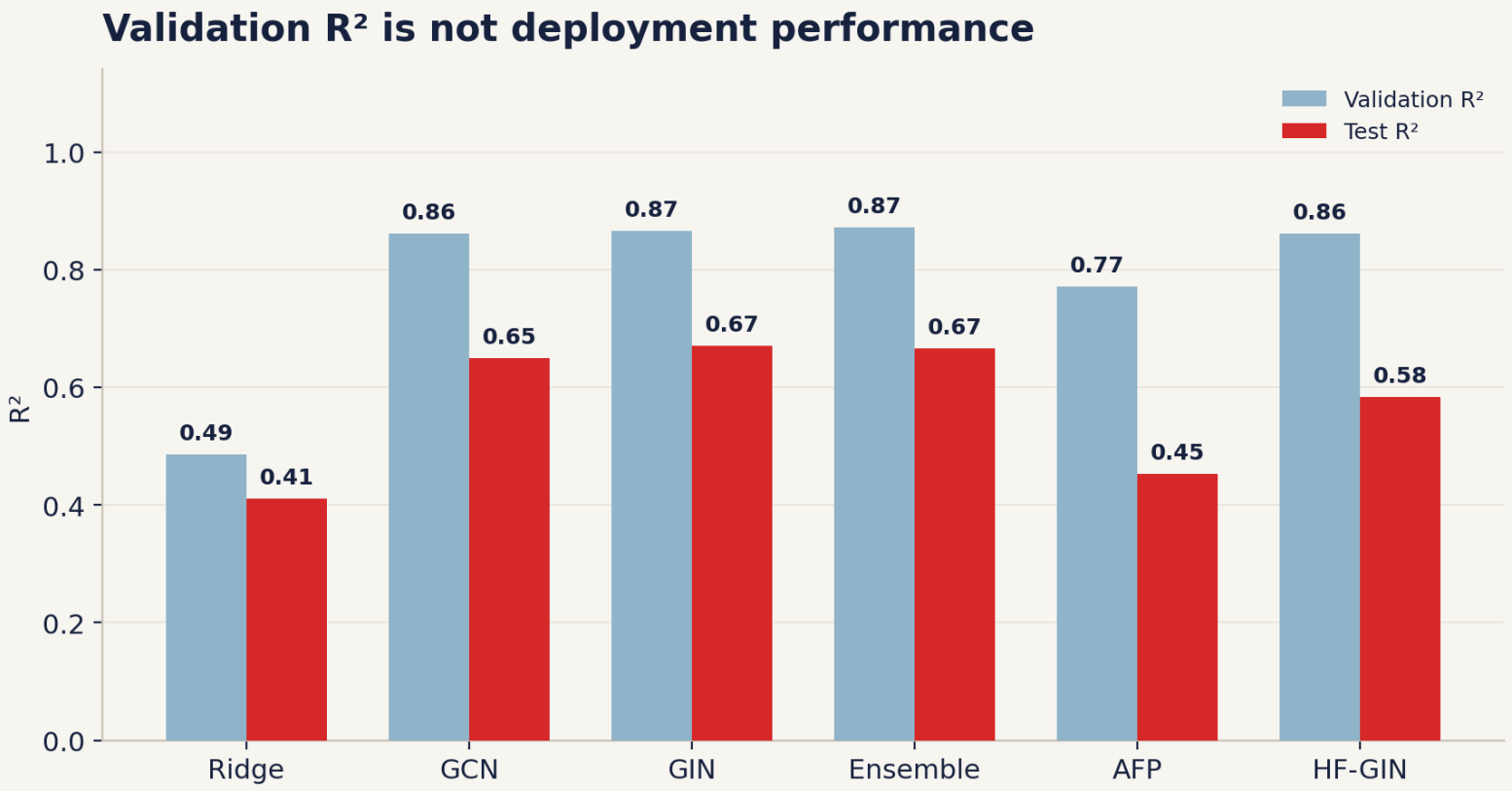
Its most extreme predictions occur exactly where evidence is weakest. This is extrapolation, not discovery.

Selection bias

Choosing the maximum of 154k noisy predictions enriches for over-predictions. This is the winner's curse.

Validation looked good because it was analog-rich

The test is scaffold-disjoint — good — but 97.7% of test compounds still pass the 0.40 similarity cut.



- 42 of 52 validation compounds share a scaffold with training.
- 30 scaffolds are shared between train and validation.
- Train–test and validation–test scaffold overlap is zero.
- Val $R^2 \approx 0.87$; test $R^2 \approx 0.65$.
- The test is cleaner than validation, but it is not the library.

Model selection, early stopping, and calibration were all chosen on the analog-rich validation set.

THE CAUSAL CHAIN

One mismatch, amplified by one selection rule

645 analogs
small, biased chemical neighborhood

→

Analog validation
42/52 val share a train scaffold

→

Select + calibrate
choose the best on analogs

→

154k library
99.4% outside AD 0.40

→

Top-100
all outside AD; mean 0.186

Two red links: the validation does not represent the library, and the shortlist selects the extrapolation tail.

Code-level amplifiers (not the root cause)

Active evidence
all 645 actives used, including test; 52 test actives self-match

NNLS label
'NNLS ensemble' row is actually the simple mean

Tier mismatch
tier uses 0.40 while the shortlist uses 0.52

Stale cache
154,587 rows vs 154,647 canonical; final shortlist missing

THE PROPER WAY

Train for the chemistry you will actually screen

A protocol that matches the deployment question, not a random benchmark.

1 Define the domain

Is the goal analog optimization or diverse-library discovery? They need different data and metrics.

2 Split by scaffold

GroupShuffleSplit / GroupKFold by Murcko scaffold. Never let analogs leak into validation.

3 Nested CV

Select and calibrate inside the training folds. Keep the test set frozen.

4 Freeze test

Compute test predictions once, at the end. No repeated peeking.

5 Train-only evidence

Active support and AD reference must use training data for held-out evaluation.

6 Report uncertainty

AD coverage, calibration slope, prediction intervals, enrichment with bootstrap CIs.

7 Prospective assay

A computational shortlist is a hypothesis. Only a wet-lab assay makes it a hit.

If the goal is a diverse library: use a classifier for enrichment + AD, then assay. Do not rank pIC50 outside the domain.

A lower, honest score is better than a high score that does not transfer.

WHAT I WOULD FIX FIRST

P0 is not a new model. P0 is an honest evaluation.

P0 — before any more claims

- Scaffold-grouped validation
- Train-only active evidence
- Freeze test predictions
- Fix/relabel NNLS result
- Align tier with 0.52
- Rebuild cache with content hashes
- Rerun the final pipeline

P1 — robustness

- Deterministic seeds / generators
- Parent + salt + stereoisomer collapse
- Explicit model availability
- Nested CV for every compared model
- Bootstrap CIs for enrichment
- Per-compound uncertainty

P2 — engineering

- Split the 2,423-line script into modules
- Add smoke tests and a pinned environment
- One config source + schema version
- Pin data by SHA256, not ‘main’
- Remove silent exception swallowing

Expected result

Lower validation/test numbers.
A smaller — possibly empty — shortlist.
That is success, not failure: it tells you what the data can support.

Do not start by tuning the model. Start by making the evaluation match the screening question.

BOTTOM LINE

A model is not a microscope.
It cannot see chemistry it was never trained on.

Your pipeline is a potentially useful analog-ranking tool, once validation is fixed. It is not yet a discovery engine for a diverse library.

The correct answer may be: “No reliable hits in this library.”

Fix the validation contract first. Then the model has a chance to be believed.